

Optimal Bidding Strategy of a Plug-In Electric Vehicle Aggregator in Day-Ahead Electricity Markets Under Uncertainty

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Abstract—With a large-scale introduction of plug-in electric vehicles (PEVs), a new entity, the PEV fleet aggregator, is expected to be responsible for managing the charging of, and for purchasing electricity for, the vehicles. We approach the problem of an aggregator bidding into the day-ahead electricity market with the objective of minimizing charging costs while satisfying the PEVs' flexible demand. The aggregator places demand bids only (no vehicle-to-grid is considered). The aggregator is assumed to potentially influence market prices, in contrast to what is commonly found in the literature. Specifically, the bidding strategy of the aggregator is formulated as a bilevel problem, which is implemented as a mixed-integer linear program. The upper level problem represents the charging cost minimization of the aggregator, whereas the lower level problem represents the market clearing. Since the bids of other market participants are not known to the aggregator *ex ante*, a simple strategy is used to estimate them, based on historical data. An aggregated representation of the PEV end-use requirements as a virtual battery, with time varying power and energy constraints, is proposed, derived from individual driving patterns. Since driving patterns cannot be perfectly forecasted, we introduce a probabilistic virtual battery model. We compare the results of the proposed bidding strategy with those of a bidding strategy assuming exogenous prices, uncontrolled charging, and a central dispatch of the fleet. We also explore the impacts of different sources of uncertainty. Results show that with the proposed bidding strategy, costs are significantly lower than under inflexible charging and are lower than assuming exogenous prices. Moreover, the approach also performs well under uncertainty. Results also suggest that the aggregator only has limited market power potential at moderate PEV penetrations.

Index Terms—Aggregator, bilevel optimization, demand side management, electric vehicles, mathematical problem with equilibrium constraints (MPEC), smart-charging, stochastic optimization, strategic bidding.

NOTATION

Constants:

Δt	Time step duration.
η	Charging efficiency of PEV aggregation.
E_{dep}^t	Energy drop from vehicles departing at time t .
E_{arr}^t	Energy contribution of vehicles arriving at time t .

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$P_{c,\text{max}}^t$	Upper bound of aggregated charging power at time t .
$P_{c,\text{min}}^t$	Lower bound of aggregated charging power at time t .
ϵ	Constraint violation parameter.
β	Confidence parameter.
n_m	Number of stochastic constraints.
$b_{s_m}^t$	Price of supply bid s_m at time t .
$b_{d_k}^t$	Price of demand bid d_k at time t .
L	Very large number.
$u_{v_l}^t$	Indicates if vehicle v_l is connected at time t (0 if not connected, 1 if connected).

Variables:

E^t	Energy content of virtual battery at time t .
P_a^t	Accepted volume of aggregator's demand bid at time t .
$\tau_{j,\text{min}}$	Lower bound of stochastic component of inequality j .
$\tau_{j,\text{max}}$	Upper bound of stochastic component of inequality j .
B_q	Hyper-rectangle confining uncertainty.
N	Number of samples of stochastic variable.
p^t	Market clearing price at time t .
b_a^t	Price of aggregator's demand bid at time t .
$P_{s_m}^t$	Accepted volume of supply bid s_m at time t .
$P_{d_k}^t$	Accepted volume of demand bid d_k at time t .
Φ	Vector of optimization variables.
$E_{v_l}^t$	Energy content of the battery of vehicle v_l at time t .
$P_{v_l}^t$	Charging demand of vehicle v_l at time t .

The upper/lower bounds of variables are indicated by the subscript max/min.

Indices:

t	Time steps $t \in \{1, \dots, T\}$, where $T = 24$.
s_m	Supply bids $s_m \in \{s_1, \dots, s_M\}$.
d_k	Demand bids $d_k \in \{d_1, \dots, d_K\}$.
v_l	Vehicles $v_l \in \{v_1, \dots, v_L\}$.
j	Stochastic constraints $j \in \{1, \dots, n_m\}$.
k	Stochastic samples $k \in \{1, \dots, N\}$.

Random variables:

- $\omega_{[y]}^t$ Error in forecast of $[y]^t$.
 δ Vector of random variables at vehicle level.

where $[y]$ is a placeholder for: E_{arr} , E_{dep} , E_{min} , E_{max} , $P_{c,\text{min}}$, and $P_{c,\text{max}}$.

Dual variables:

The dual variables below are associated with the following constraints:

- λ^t Supply-demand equilibrium.
 μ_{min,m_i}^t Bid m_i volume lower bound.
 μ_{max,m_i}^t Bid m_i volume upper bound.

Binary variables:

The binary variables below are associated with the following constraints:

- $\gamma_{\text{min},m_i}^t$ Bid m_i volume lower bound.
 $\gamma_{\text{max},m_i}^t$ Bid m_i volume upper bound.

I. INTRODUCTION

WHEN plug-in electric vehicle (PEV) shares reach a significant level, managing their charging becomes important, both to avoid voltage and congestion problems in the grid [1]–[3], as well as from an economic perspective to reduce charging costs [3], [4]. PEV demand can be considered flexible since most vehicles are parked for a long time and most trips do not require the full capacity of the PEV battery. However, any charging management or “smart-charging” scheme should take driver needs into account in order not to disrupt vehicle end-use. An aggregator could manage the charging and purchasing of electricity on behalf of the PEV owners. By agreeing to a charging management scheme, vehicle owners would benefit from lower electricity rates for charging, since the aggregator can exploit charging flexibility to reduce the charging costs and can share part of this benefit with the PEV owners.

In this paper we focus on aggregator-based centralized charging management approaches, for decentralized approaches we refer to [5] and the references therein.

The role of the aggregator in integrating PEVs in power system structures is discussed in several publications [6]–[11]. The aggregator clusters a large number of small flexible loads/storage devices, which individually have only a small impact on the system, but as a group can be used as a significant resource. The aggregator is envisaged to directly or indirectly manage vehicle charging [11], e.g., to minimize charging costs, but also to provide ancillary services to the grid [9], such as regulating power.

In this paper we propose a strategy for a PEV aggregator that places demand bids in the day-ahead market, where we explicitly model the market clearing process. Both driver end-use constraints and their related uncertainties, as well as the uncertainty in the bids of other market participants are taken into account. This paper focuses on how the costs of purchasing energy can be minimized by the aggregator. We do not consider how this

cost reduction is shared with PEV owners. The two problems can be addressed separately; see, e.g., [12] for a mechanism to distribute profits fairly among demand side resources. The aggregator's cost minimization problem is formulated as a bilevel optimization problem where the upper level represents the minimization of charging costs, which is constrained by the available flexibility of charging demand. The lower level represents the market clearing process, where the bids of other participants are estimated based on past data. To determine the available charging flexibility, the fleet is aggregated into a virtual battery with time-varying energy and power constraints. These are modeled as probabilistic constraints to take account of driving pattern uncertainty. The constraints are parameterized using information on individual driving patterns obtained from a transport simulation.

A number of papers address the aggregator's cost optimization problem, where the costs of purchasing electricity are minimized subject to constraints on PEV demand. In many of these approaches prices are considered as an exogenous parameter, unaffected by PEV demand [13]–[16]. In this paper we show that this assumption can lead to suboptimal results, even for very low penetration rates, and therefore propose a model with endogenous price formation. In other endogenous price models, linear [17], [18] and nonlinear [18] approximations of the price as a function of the load are used. However, the relationship between load, i.e., the sum of accepted demand bids, and the market price is not simple. Whether a demand bid is accepted or not depends on the willingness to pay for electricity (demand bid price) and on the bids of the other market participants. Without modeling the market clearing process, it is difficult to determine a bidding strategy for the aggregator. Existing models focus on optimal demand *volumes* given endogenous/exogenous prices, but not on the corresponding bid *prices* to achieve these volumes. *The first contribution of the paper is to model the aggregator as a price-maker and analytically take into account the impact of the aggregator's bids on prices using a bilevel formulation.* This formulation is inspired by the model proposed in [19] for the bidding of a strategic producer. We also address the issue that the market clearing prices/market bids are not known ex ante. Only some of the mentioned approaches consider the fact that the market price is uncertain [13], [15], [18].

Another important aspect of charging optimization is the modeling of the fleet's demand. Since charging is not an exogenous input, but rather a variable managed by the aggregator, PEV demand forecasting is different to that of conventional loads. Therefore, instead of an expected demand profile, the aggregator needs to determine a set of constraints that the optimal charging profile should satisfy. Based on driving patterns, an aggregated representation of the constraints of the fleet is incorporated in the optimization problem of the aggregator, such as in [15], [17], [18], and [20]. However, individual driving patterns are subject to uncertainty. Therefore, the aggregated representation of the fleet should ideally be a probabilistic one. This has been addressed in [16] and [21], whereas only in [21] a derivation of the probabilistic properties of the aggregation based on driving patterns is provided. Extending the framework we proposed in [21], the fleet's constraints are modeled as chance constraints which are reformulated with

a scenario-based approach offering probabilistic guarantees on constraint violation [21]. *The second contribution of the paper is a model for load flexibility as a set of probabilistic time-varying power and energy constraints, constructed from samples of driving patterns, and reformulated using a scenario-based robust approach.*

We compare the proposed bidding strategy with the following benchmarks: 1) the optimal strategy assuming exogenous prices, 2) a surplus maximizing central dispatch, and 3) a case where PEV demand is considered to be inflexible (uncontrolled charging). Moreover, we analyze the impact of two sources of uncertainty: 1) uncertainty in market bids and 2) uncertainty in driving patterns. *Based on the comparison of the different approaches at varying levels of uncertainty, the third contribution of the paper is an analysis of 1) the value of information, 2) the value of flexibility, 3) the value of models with endogenous prices, 4) the aggregator's impact on prices, and 5) the market power of the aggregator.*

This paper extends the work we presented in [23] by considering PEV driving pattern uncertainty and by comparing our endogenous price model with exogenous price models.

We perform a case study with historical market data and realistic driving patterns, which shows that the proposed bidding strategy results in lower costs, compared with uncontrolled charging and the approach assuming exogenous prices. Moreover, the strategy performs well under driving pattern uncertainty and also satisfactorily under market bid uncertainty, even with a simple estimation strategy for the market bids of other participants. Furthermore, the results indicate that the aggregator has limited market power potential at moderate PEV penetrations.

The proposed method is described in Section II, and is then tested with a case study in Section III. Finally, Section IV concludes the paper.

II. METHODOLOGY

Our approach is conceptually similar to the three-step approach proposed in [20], where, as a first step, individual PEV constraints are aggregated, and then the optimal fleet schedule is determined by solving an optimization problem (second step). The third step is the real-time control step, where individual vehicles are scheduled. In this paper we cover the first step (Section II-A) and second step (Section II-B) only, see [24] for other work concerning the scheduling of individual vehicles. Because aggregated models do not perfectly represent individual requirements [25], small deviations from the aggregated schedule can be expected in the real-time dispatch. The aggregator could, e.g., take positions in intra-day markets to cope with expected deviations from its day-ahead schedule.

First, we describe the probabilistic flexible demand model as a virtual battery (Section II-A), then we formulate the bilevel problem (Section II-B) and derive a bidding strategy (Section II-C), and finally we define the benchmark approaches (Section II-D).

A. Probabilistic Flexible Demand Model

To decide on the optimal bidding strategy, the aggregator first needs to estimate the energy needs of the fleet over time, as well

as the available charging flexibility. In the following section we describe how the set of aggregate charging profiles that satisfy drivers' needs can be captured by a probabilistic time-varying battery model.

1) *Individual Driving Patterns:* Individual driver end-use constraints depend on driving patterns, i.e., on departure and arrival times and trip energy consumption. The driving patterns used in this paper stem from the transport simulation tool MATSim [26]. MATSim is an agent-based simulation where each agent has a set of activities to be performed (e.g., work, shopping, etc.) and the optimization selects the driving patterns that maximize agents' utility, taking into account factors such as the induced traffic and the available means of transportation.

In practice, the PEV drivers would need to communicate their planned trips to the aggregator in advance, or alternatively machine learning algorithms could be used to forecast driving patterns. Specifically, the aggregator needs to know the timing of the trips and the associated expected energy consumption, as well as parameters such as the battery size and the maximum charging power.

2) *Flexible Demand as Time-Varying Battery:* For large fleets, it is not computationally tractable to simultaneously optimize the charging of each individual vehicle. Therefore, an aggregation model is needed to condense driving pattern information into a form that can be used in the aggregator's cost optimization. Here, the fleet is modeled as a virtual battery clustering the batteries of PEVs connected to the grid at each point in time. Thus, the energy capacity and available charging power of the virtual battery change over time.

The virtual battery is described by the following equations:

1) Battery dynamics:

$$E^t = E^{t-1} + P_a^t \cdot \eta \cdot \Delta t + E_{\text{arr}}^t + \omega_{E_{\text{arr}}}^t - \left(E_{\text{dep}}^t + \omega_{E_{\text{dep}}}^t \right) \quad \forall t. \quad (1)$$

2) Battery constraints:

$$E^0 = E^T \quad (2)$$

$$E_{\min}^t + \omega_{E_{\min}}^t \leq E^t \leq E_{\max}^t + \omega_{E_{\max}}^t \quad \forall t \quad (3)$$

$$P_{c,\min}^t + \omega_{P_{c,\min}}^t \leq P_a^t \leq P_{c,\max}^t + \omega_{P_{c,\max}}^t \quad \forall t. \quad (4)$$

The evolution of the energy content of the virtual battery, which represents the sum of the individual energy contents of connected PEVs, over time depends on the aggregated charging power and on the arrivals and departures of PEVs, as shown in (1). Here, E_{arr}^t represents the increase in energy content due to the arrival of vehicles at a given time step, and E_{dep}^t represents the reduction in energy content due to the PEV departures. The first constraint (2) ensures that enough energy is purchased, i.e., the energy content at the end of the simulation is equal to the energy content at the beginning of the simulation. This constraint is included to avoid myopic behavior: If it were omitted, because of cost minimization, the solution would be to buy as little energy as possible for the day considered in the optimization and to shift demand to a time beyond this horizon. Equation (2) thus makes sure that the aggregator buys enough energy

to cover the daily energy need of the PEVs.¹ Inequality constraints (3) and (4) enforce the upper and lower bounds of the energy content and charging power of the virtual battery. The lower energy bound constitutes an aggregated representation of the charging deadlines of the PEVs. It expresses how much energy should be in the virtual battery at a particular time step and therefore specifies at least how much energy should have been purchased by then.

The aggregated parameters describing the virtual battery, i.e., $E_{\text{arr}}, E_{\text{dep}}, E_{\text{min}}, E_{\text{max}}, P_{c,\text{min}}$ and $P_{c,\text{max}}$, are obtained from the individual driving patterns as explained in Appendix A-A. These parameters are subject to uncertainty, since driving behavior cannot be perfectly forecasted. Therefore all the variables which are derived from the aggregation of individual driving patterns are subject to a forecasting error denoted ω .

3) *Reformulation of Stochastic Inequality Constraint:* To ease further explanations, inequality constraints (3) and (4), where E^t in (3) has been replaced by (1), $E_{\text{min}}^t \leq E^0 + \sum_{\tau=1}^t (P_a^\tau \cdot \eta \cdot \Delta t + E_{\text{arr}}^\tau + \omega_{E_{\text{arr}}}^\tau - E_{\text{dep}}^\tau - \omega_{E_{\text{dep}}}^\tau) \leq E_{\text{max}}^t$, are rewritten as generic linear constraints with an uncertain parameter ω_j :

$$a'_j \cdot x + \omega_j \leq b_j \quad j = 1, \dots, n_m. \quad (5)$$

The number of inequality constraints, n_m , is thus $4T$, and $x := [E_0, P_a^1, \dots, P_a^T]$. The random variables ω_j represent the uncertainty associated with the aggregated fleet representation, which is a function of the vector of random variables δ at the individual vehicle level, such as departure times, trip durations and trip energy consumptions:

$$\omega_j = q_j(\delta). \quad (6)$$

The constraints in (5) are formulated as a joint chance constraint, i.e., they are to be jointly satisfied with a given probability $1 - \epsilon$:

$$\mathbb{P} \left(\delta \in \Delta \mid \max_{j=1, \dots, n_m} a'_j \cdot x + q_j(\delta) - b_j \leq 0 \right) \geq 1 - \epsilon. \quad (7)$$

We use the approach proposed in [22] and [27] to solve the joint chance constraint. The general idea is that we can solve a robust counterpart of (7) where the uncertainty $q_j(\delta)$ is confined in a hyper-rectangle B_q^* . The set B_q^* is obtained by solving a random program where a finite number of samples of δ is extracted:

$$\min_{\tau_{j,\text{max}}, \tau_{j,\text{min}}, \forall j} \sum_j (\tau_{j,\text{max}} - \tau_{j,\text{min}}) \quad (8)$$

$$\text{s.t. } q_j(\delta^k) \in [\tau_{j,\text{min}}, \tau_{j,\text{max}}] \quad \forall j, \forall k = 1, \dots, N. \quad (9)$$

The hyperrectangle B_q^* is determined using the Cartesian product of the resulting intervals $[\tau_{j,\text{min}}^*, \tau_{j,\text{max}}^*]$ as $B_q^* := \times_{j=1}^{n_m} [\tau_{j,\text{min}}^*, \tau_{j,\text{max}}^*]$. The number of required samples N depends on the number of uncertainty functions $q_j(\delta)$, i.e., n_m , according to the following equation:

$$N \geq \left\lceil \frac{1}{\epsilon} \frac{e}{e-1} \left(2n_m - 1 + \ln \frac{1}{\beta} \right) \right\rceil. \quad (10)$$

The confidence parameter β represents the probability of constraint (7) not holding because the extracted samples are not representative. It can be set to be very low, since it only affects the number of required samples logarithmically. Note that, although the dimension of δ is very large (several stochastic variables per vehicle), this does not affect the number of scenarios to be extracted. The generation of random samples of driving patterns is explained in Appendix A-B.

Finally, with the definition $q(\delta) := [q_1(\delta), \dots, q_{n_m}(\delta)]$, (7) can be formulated as

$$\max_{j=1, \dots, n_m} \max_{q(\delta) \in B_q^* \cap q(\Delta)} a'_j \cdot x + q_j(\delta) - b_j \leq 0. \quad (11)$$

Because of the structure of the constraints, this can be further simplified to

$$a'_j \cdot x \leq b_j - \tau_{j,\text{max}}^* \quad j = 1, \dots, n_m. \quad (12)$$

In conclusion, by considering N possible scenarios of fleet driving pattern configurations, the inequality constraint is tightened by a margin $\tau_{j,\text{max}}^*$.

The advantage of the method [22] used to reformulate the joint chance constraint is that it does not rely on any particular assumption about the distribution of the random variables ω_j . It is only necessary to be able to extract samples of these variables, which is achieved by generating different samples of driving patterns for each vehicle. For this purpose departure and arrival times, and trip energy consumptions are modeled as random variables, represented by the vector of random variables δ . For each of the individual samples, the corresponding values of the aggregated parameters $E_{\text{arr}}, E_{\text{dep}}, E_{\text{min}}, E_{\text{max}}, P_{c,\text{min}}$, and $P_{c,\text{max}}$ are computed. The values of ω_j are therefore not directly sampled, but obtained indirectly from the aggregation of individual vehicle samples, as represented by (6). More details on the sampling and aggregation procedure are given in the Appendix A.

4) *Reformulation of Stochastic Equality Constraint:* With uncertainty in the evolution of the energy content E^t , the equality constraint (2) cannot be satisfied for several possible outcomes of the uncertain variables. By substituting (1) into (2)

$$\begin{aligned} E^T &= E^0 + \sum_t \left(P_a^t \cdot \eta \cdot \Delta t + E_{\text{arr}}^t + \omega_{E_{\text{arr}}}^t - E_{\text{dep}}^t - \omega_{E_{\text{dep}}}^t \right) \\ &= E^0 \end{aligned} \quad (13)$$

and using the median to minimize the mean absolute deviations from constraint (2), (2) is reformulated as

$$\sum_t (P_a^t \cdot \eta \cdot \Delta t + E_{\text{arr}}^t - E_{\text{dep}}^t) = \text{median} \left(\sum_t (\omega_{E_{\text{dep}}}^t - \omega_{E_{\text{arr}}}^t) \right). \quad (14)$$

The median is estimated out of the N samples extracted to solve the inequality constraints.

¹Alternatively to constraint (2), a value could be assigned to the energy at the end of the time horizon in the objective function. Constraint (2) can lead to suboptimal results when driving patterns change significantly from one day to another. For illustration purposes, we have chosen to fix the daily energy to be purchased by the aggregator to be able to better compare different approaches, see Section II-D.

B. Bilevel Model

The goal of the aggregator is to find a bidding strategy for the day-ahead market that minimizes the cost of purchasing the charging energy while satisfying PEV driver end-use constraints, modeled as constraints on a virtual battery. The electricity price is the outcome of a market clearing process, based on the aggregator's demand bids and the other market participants' supply and demand bids. The aggregator's bidding strategy can be modeled as a bilevel problem where the upper level problem corresponds to the aggregator's cost minimization, while the lower level problem corresponds to the market clearing process, i.e., the maximization of supplier and consumer surplus. Mathematically, the bilevel problem corresponds to a mathematical problem with equilibrium constraints (MPEC). In this type of approach, the reaction of other market participants to the aggregator's bids is not explicitly modeled. Since the aggregator's demand is flexible, it is assumed that the bids of other participants will not change significantly as a response to the aggregator's demand, i.e., other agents cannot easily exercise market power against the aggregator. The upper level is formulated as follows:

$$\min_{E^0, b_a^t, \forall t} \sum_t p^t \cdot P_a^t \quad (15)$$

subject to the linear constraints (12) and (14) and

$$p^t = \lambda^t \quad (16)$$

where the accepted bid volume P_a^t and market clearing price p^t stem from the solution of the lower level problem:

$$\min_{P_{s_m}^t, P_{d_k}^t, P_a^t} \sum_{s_m, t} b_{s_m}^t \cdot P_{s_m}^t - \sum_{d_k, t} b_{d_k}^t \cdot P_{d_k}^t - \sum_t b_a^t \cdot P_a^t \quad (17)$$

subject to

$$P_{s_m}^t = P_{d_k}^t + P_a^t : \lambda^t \quad \forall t \quad (18)$$

$$0 \leq P_{m_i}^t \leq P_{m_i, \max}^t : \mu_{\min, m_i}^t, \mu_{\max, m_i}^t \quad \forall t, \forall m_i \in M. \quad (19)$$

In the lower level problem, the total surplus of the market participants is maximized (17), given the supply and demand equilibrium (18) and bid volumes (19). The index m_i groups the indices of the demand, supply and aggregator bids with $M = \{s_1, \dots, s_M, d_1, \dots, d_K, a\}$. The Lagrange multipliers associated with each constraint are introduced after the colon. The Lagrange multiplier of the supply-demand equilibrium constraint (18) represents the market clearing price, as seen by the aggregator in the upper level problem (16).

In this paper the energy content of the virtual battery at time zero, E^0 , is considered an endogenous variable. As the market clearing of the day ahead market takes place hours before $t = 0$ actually occurs, the aggregator has the possibility of driving the energy content to the desired value up to that time. Note that because $E^0 = E^T$ and E^T being bounded through equation (3), E^0 cannot take arbitrary values. We choose E^0 to be endogenous to keep results as general as possible. However, the proposed problem formulation is also valid for a fixed E_0 .

The lower level problem is linear, and therefore can be replaced by its Karush-Kuhn-Tucker conditions. This leads to the following additional constraints:

$$b_{s_m}^t - \lambda^t - \mu_{\min, s_m}^t + \mu_{\max, s_m}^t = 0 \quad \forall t, \forall s_m \quad (20)$$

$$-b_{d_k}^t + \lambda^t - \mu_{\min, d_k}^t + \mu_{\max, d_k}^t = 0 \quad \forall t, \forall d_k \quad (21)$$

$$-b_a^t + \lambda^t - \mu_{\min, a}^t + \mu_{\max, a}^t = 0 \quad \forall t \quad (22)$$

$$0 \leq P_{m_i}^t \perp \mu_{\min, m_i}^t \geq 0 \quad \forall t, \forall m_i \in M \quad (23)$$

$$0 \leq P_{m_i, \max}^t - P_{m_i}^t \perp \mu_{\max, m_i}^t \geq 0 \quad \forall t, \forall m_i \in M. \quad (24)$$

The nonlinearities associated with the complementarity slackness conditions (23), (24) can be linearized with the introduction of integer variables, denoted γ , and very large constants, denoted L , as described in [19]:

$$P_{m_i}^t \leq (1 - \gamma_{\min, m_i}^t) \cdot L_{P_{\min, m_i}^t} \quad \forall t, \forall m_i \in M \quad (25)$$

$$\mu_{\min, m_i}^t \leq \gamma_{\min, m_i}^t \cdot L_{\mu_{\min, m_i}^t} \quad \forall t, \forall m_i \in M \quad (26)$$

$$P_{m_i, \max}^t - P_{m_i}^t \leq (1 - \gamma_{\max, m_i}^t) \cdot L_{P_{\max, m_i}^t} \quad \forall t, \forall m_i \in M \quad (27)$$

$$\mu_{\max, m_i}^t \leq \gamma_{\max, m_i}^t \cdot L_{\mu_{\max, m_i}^t} \quad \forall t, \forall m_i \in M. \quad (28)$$

Finally, the objective of the upper level problem (15) can be linearized using the strong duality theorem. For further details on this procedure we refer to [19]:

$$\begin{aligned} \sum_t p^t \cdot P_a^t &= \sum_{s_m, t} b_{s_m}^t \cdot P_{s_m}^t - \sum_{d_k, t} b_{d_k}^t \cdot P_{d_k}^t \\ &\quad + \sum_{s_m, t} \mu_{\max, s_m}^t \cdot P_{\max, s_m}^t \\ &\quad + \sum_{d_k, t} \mu_{\max, d_k}^t \cdot P_{d_k, \max}^t. \end{aligned} \quad (29)$$

In summary, the problem can be formulated as the following mixed integer linear programming problem:

$$\min_{\Phi} \left\{ \sum_{s_m, t} b_{s_m}^t \cdot P_{s_m}^t - \sum_{d_k, t} b_{d_k}^t \cdot P_{d_k}^t + \sum_{s_m, t} \mu_{\max, s_m}^t \cdot P_{\max, s_m}^t + \sum_{d_k, t} \mu_{\max, d_k}^t \cdot P_{d_k, \max}^t \right\} \quad (30)$$

with the optimization variables

$$\Phi = [E^0, b_a^t, P_{m_i}^t, \lambda^t, \mu_{\min, m_i}^t, \mu_{\max, m_i}^t, \gamma_{\min, m_i}^t, \gamma_{\max, m_i}^t] \quad (31)$$

subject to the constraints (12) and (14), (18)–(22), (25)–(28) and

$$\mu_{\min, m_i}^t, \mu_{\max, m_i}^t \in \mathbb{R}^+ \quad \forall m_i \in M \quad (32)$$

$$\gamma_{\min, m_i}^t, \gamma_{\max, m_i}^t \in \{0, 1\} \quad \forall m_i \in M. \quad (33)$$

C. Bidding Strategy

To be able to apply the described bilevel optimization, the aggregator needs to estimate the bids of the other market partic-

ipants as they are not known by the aggregator before it places its own bids. In this paper we analyze a simple strategy where the aggregator uses the bids from the days of the previous week as the best estimate for the bids of the days of the current week. For example, for a given Monday at noon, the bids of the previous Monday at noon are used. This information is commercially available at power exchanges such as the European Energy Exchange in the form of aggregated price curves.

The optimization described in the previous subsection generates two important results:

- the accepted aggregator bid volumes P_a^{t*} , which represent the optimal aggregated charging profile;
- the optimal aggregator bid prices b_a^{t*} , which are in all cases equal to the clearing prices λ^{t*} [19].

The most straightforward way to define the aggregator's bid volumes and prices, would be to choose 1) the upper bound $P_{c,\max}^t$, or more precisely its robust counterpart according to (12), as bid volume, and 2) the bid price resulting from the optimization b_a^{t*} as bid price. Due to uncertainties regarding the market bids and the corresponding clearing prices, using that strategy would usually mean that either excess or insufficient energy is purchased, even if the market prices are close to the estimated ones. If, at a given time step, the actual market price is only marginally lower than the expected price, the bid will not be accepted. On the other hand, if the market clearing price is slightly higher, the accepted bid volume could be higher than the expected volume.

To avoid this problem, we use the accepted bid volumes P_a^{t*} as bid volumes and add a markup on the bid price obtained from the bilevel optimization. This option was chosen because the accepted bid volumes satisfy the power and energy constraints of the virtual battery (12) and (14), and at the same time indicate at which time it is probably less costly to purchase energy. A more detailed discussion on the implications of the price markup choice is given in Section III-C.

D. Benchmarks

The results obtained with the described bidding strategy under different levels of uncertainty are compared with other theoretical benchmark cases.

Reference without PEV: Reference market clearing without the participation of the PEV aggregator.

Exogenous prices: Here the aggregator considers itself a price-taker when deciding on its bids. Therefore its optimization problem is equivalent to the upper level problem of the presented bilevel model. The lower level is not needed since the aggregator assumes its bidding decisions do not affect market prices.

Central dispatch: Here the charging profile is obtained which maximizes supplier and consumer surplus of the existing market participants while satisfying the constraints of the PEV aggregation, that is

$$\min_{P_{s_m}^t, P_{d_k}^t, P_a^t} \sum_{s_m, t} b_{s_m}^t \cdot P_{s_m}^t - \sum_{d_k, t} b_{d_k}^t \cdot P_{d_k}^t \quad (34)$$

subject to (12)–(14) and (18), (19). This theoretical case represents the optimal use of charging flexibility from a system perspective.

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Inflexible PEV demand: In this case we assume that charging is left uncontrolled, i.e., each PEV starts charging as soon as it is parked, and calculate the resulting aggregated charging profile. The aggregator has to purchase this energy in the market, independent of the price. This is equivalent to a bid with the maximum allowed bid price and the aggregated charging demand at each time step as the bid volume.

The different approaches can be analyzed for different levels of uncertainty. Under perfect information, both driving patterns and market bids are forecasted perfectly. However, in practice the aggregator faces uncertainty in market bids and uncertainty in driving patterns. If driving patterns are perfectly forecasted, then $\tau_{j,\max}^*$ in (12), and $\omega_{E_{dep}}^t$ and $\omega_{E_{arr}}^t$ in (14) are zero. As mentioned previously, if the market bids are not known to the aggregator ex ante, its simple estimate is that they are equal to those of the previous week. In the case of exogenous prices, the price expectations are also formed on the same basis. Given the bids computed with these price expectations, the final cost outcome is computed using the actual realization from the time series.

III. CASE STUDY

A. Case Study Setup

To validate the proposed approach we simulate the bidding strategy and benchmark cases with aggregated price curves of the spot market for electricity for the bidding area Germany/Austria, obtained from the European Energy Exchange (EEX). The simulation period spans the year 2012. Each day is divided into hourly time steps, that is Δt corresponds to one hour. We assume a 2% penetration of PEVs in Germany and Austria, which corresponds to almost one million PEVs. The value of 2% is chosen because it is a number low enough to be realistic in the near future, but already high enough to affect market prices, as will be shown later. In Germany, the goal is to have 1 million PEVs by 2020 [28].

The driving patterns used in the case study, obtained from the tool MATSim, are specific to Switzerland and represent typical mobility behaviour on weekdays. With this transport data it is therefore not possible to make a quantitative analysis of the potential impact of a PEV aggregator on the German/Austrian electricity spot market. However, no equivalent simulations exist for Germany and Austria and for weekend mobility. Nevertheless, it is still possible to qualitatively test the performance of the described bidding strategy, which is the goal of this paper. The assumed maximum charging power for vehicles is 3.7 kW and the charging efficiency 90%. The energy consumption is derived from trip distances with the factor 0.2 kWh/km, which results in an average electricity consumption per day and PEV of 5.8 kWh, with standard deviation 7.3 kWh. Battery capacities of 16 kWh and 24 kWh are used, each for half of the fleet. For the probabilistic constraints, the violation parameter ϵ was set to 0.08 and the confidence parameter β to $1e^{-7}$, meaning that almost 2290 samples of driving patterns need to be extracted for each vehicle.

B. Cost Comparison

First, we compare the costs of the bidding strategy with those of the benchmark cases under different sources of uncertainty.

TABLE I
COST COMPARISON-AVERAGE DAILY COST OF PURCHASING ENERGY
NORMALIZED BY DAILY ENERGY DEMAND [€/MWh]

	Bidding strategy	Exogenous prices	Central dispatch	Inflexible
perfect information	32.55	35.56	33.18	48.50
bid uncertainty	35.03	37.52	-	-
driving uncertainty	33.11	36.97	33.78	-
both uncertainties	35.49	38.83	-	-

Average daily costs per MWh are shown in Table I for the simulated year 2012. The results of the bidding strategy reported in this table are calculated using the maximum bid price as the bid price. Thereby it is guaranteed that in each of the approaches the same daily amount of energy is purchased, making the results of the different approaches directly comparable. A more detailed discussion on less conservative choices for the price markup added to the price obtained from the bilevel model follows in the next subsection.

For any level of uncertainty, the proposed bidding strategy performs better than under the assumption of exogenous prices. As will be shown later, this is due to the fact that if the aggregator does not take its impact on prices into account, it tends to concentrate demand too much on a few hours with expected low prices. Moreover, the bidding strategy under perfect information achieves costs only slightly lower than the central dispatch. This means that the aggregator could exercise market power by choosing a strategy slightly less costly for it but which slightly reduces the surplus of other market participants. However, the difference is very small. The intuition behind this result is that the aggregator tries to increase prices as little as possible to minimize its costs, which is also beneficial for the system.

The comparison between the scenarios with perfect information and those under uncertainty reveals the value of information. The central dispatcher never faces uncertainty in market bids when dispatching the vehicles, since the fleet is dispatched within the market clearing. With market bid uncertainty, the aggregator incurs higher charging costs than if the fleet would be dispatched centrally. Therefore, in practice, the aggregator cannot do better than a central dispatcher because of imperfect information. Driving uncertainty has a minor impact on prices under the bidding strategy, since given the large number of vehicles in the fleet, the constraints of the virtual battery can be predicted quite accurately, as explained below. Uncertainty in driving patterns could become more important with small fleet sizes, e.g., less than 1000 PEVs. Here, the impact of market bid uncertainty is more significant than that of driving pattern uncertainty. Nevertheless, the performance of the simple bid estimation strategy is still satisfactory. If a more sophisticated method is used, the results could become even closer to the perfect forecast benchmark, possibly enabling the aggregator to effectively exercise market power. Finally, the uncontrolled charging strategy (inflexible demand) leads to much higher costs than any of the other strategies (flexible demand), which confirms the significant value of flexibility.

A more detailed assessment of the impact of uncertainty on costs is given in Table II. Here the statistics of the difference

TABLE II
IMPACT OF UNCERTAINTY ON COSTS FOR THE BIDDING STRATEGY-DIFFERENCE
BETWEEN THE DAILY NORMALIZED COSTS WITH PERFECT INFORMATION
AND WITH UNCERTAINTY [€/MWh]

	min	mean	max	s.d.
bid uncertainty	0.15	2.48	40.07	3.37
driving uncertainty	0.18	0.57	4.59	0.33
both uncertainties	0.47	2.49	43.93	3.52

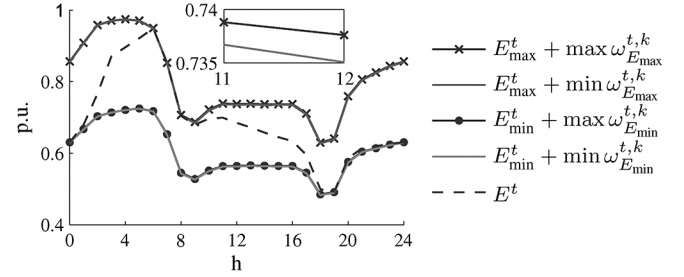


Fig. 1. Upper and lower bounds of the virtual battery's energy content and exemplary energy profile, and zoom on the upper bound between 11–12 h. The values are normalized by total fleet capacity.

between the daily normalized per MWh costs with the bidding strategy under different sources of uncertainty and the bidding strategy with perfect information across the different days of the year are reported. The costs under uncertainty are typically not much higher than with perfect information, but under bid uncertainty, in the worst case, costs can be much higher. Extreme outcomes could however be mitigated by choosing a less conservative price markup strategy.

The bounds of the virtual battery's energy content on a weekday, as well as an exemplary energy content profile, are shown in Fig. 1. The bounds are shown for the highest and lowest values of their associated forecast error among the extracted samples. The forecast errors are very small, making the difference between the two values of the bound only noticeable when zooming in. This explains the minor impact of driving pattern uncertainty on costs. For smaller fleets, the difference could be much larger, as we showed in [25]. In Fig. 1 it can also be seen that two dips in the bounds occur around 8 a.m. and 6 p.m., mainly due to commuting to/from work and that most of the vehicles are parked between 2 a.m. and 6 a.m.

C. Impact of Price Markup

In Section II-C we mentioned that a markup should be applied on the prices obtained from the bilevel model. The results discussed so far are based on the maximum possible markup, by setting the maximum bid price as bid price. In practice the aggregator would probably not be so conservative and would set a lower markup, at the risk of not purchasing sufficient energy in the day-ahead market. When this shortfall occurs the missing energy needs to be purchased elsewhere, e.g., in the intra-day market. Table III gives an overview of the trade-off between the average cost per unit of energy purchased and the percentage of energy demand covered in average in the case of market bid uncertainty. These results motivate the need for a

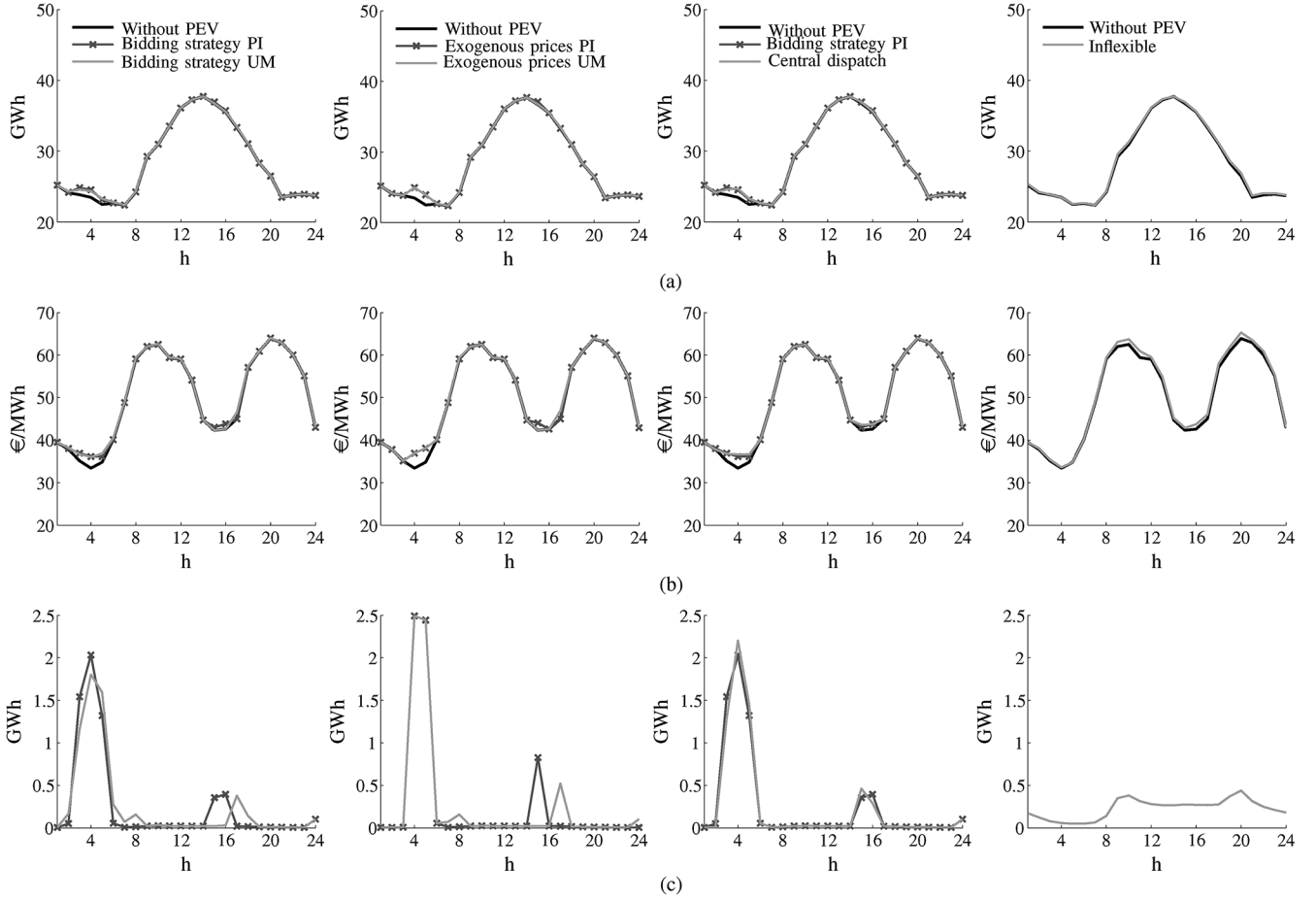


Fig. 2. Profiles for Wednesday September 6th, 2012. PI: perfect information, UM: market bid uncertainty. The first and second columns show the results of the bidding strategy and the exogenous prices benchmark, respectively, with and without market bid uncertainty. The third column illustrates the difference between the bidding strategy with perfect information and the central dispatch. Finally, column four depicts the results with inflexible demand. (a) Traded volumes. (b) Market prices. (c) Charge profiles.

TABLE III
IMPACT OF PRICE MARKUP ON COSTS AND PURCHASED ENERGY

Markup [€]	0	5	10	15
Average cost [€/MWh]	30.79	33.44	34.39	34.69
Covered demand [%]	48	74	85	91

markup since in its absence less than 50% of demand is covered. Depending on the expected cost of purchasing the additional energy through means other than the day-ahead market, the aggregator can choose an appropriate markup.

D. Typical Patterns

Further, we analyze some typical patterns in the results. Fig. 2 shows hourly trading volumes and spot prices, as well as the energy purchased by the aggregator for different approaches during a particular day. At the simulated penetration rate of 2%, PEV demand only represents a small percentage of the traded volume (less than 1%), however the impact on prices is clearly visible, explaining why assuming exogenous prices leads to suboptimal results. Fig. 2(c) (columns 1 and 2) shows that under the assumption of exogenous prices, the aggregator will concentrate charging even more than under the proposed bidding strategy. This increases prices during low-price hours

to a higher than optimal amount, as seen in Fig. 2(b) (columns 1 and 2). For all cases where PEV demand is flexible, most PEV demand is shifted to the night, when prices are low, and also some charging takes place between the two daily price peaks, see Fig. 2(b) (columns 1 to 3). The latter cannot be avoided since there is not enough flexibility to shift all demand to the less expensive night hours due to end-use constraints. By comparing between the aggregator demand with perfect information and under bid uncertainty (Fig. 2(c), columns 1 and 2), both for the bidding strategy and the exogenous price model strategy, it is clear that the profiles have a similar structure, with a large peak during night and a smaller one during the day. However, due to the imperfect forecast of the market bids/prices under bid uncertainty, the profiles deviate from the perfect information profiles. In this particular case we can see that the aggregator expects the low prices during shoulder hours to come later, so the second charging peak is shifted to the right compared with the one under perfect information. Fig. 2(b) (column 3) shows that the price profile of the central dispatch has a valley-filling type of structure. At the time periods with PEV demand, prices are approximately constant. The price profile of the bidding strategy under perfect information deviates slightly from this profile, managing to have a smaller impact on prices at the times when most of the charging is scheduled. Finally,

TABLE IV
PROBLEM SIZE AND EXECUTION TIME (MILP)

	min	mean	max
# of variables (total)	14 434	25 436	41 869
# of integer variables	5 754	10 155	16 728
# of equality constraints	2 902	5 102	8 389
# of inequality constraints	11 604	20 405	33 552
execution time [s]	17	44	169

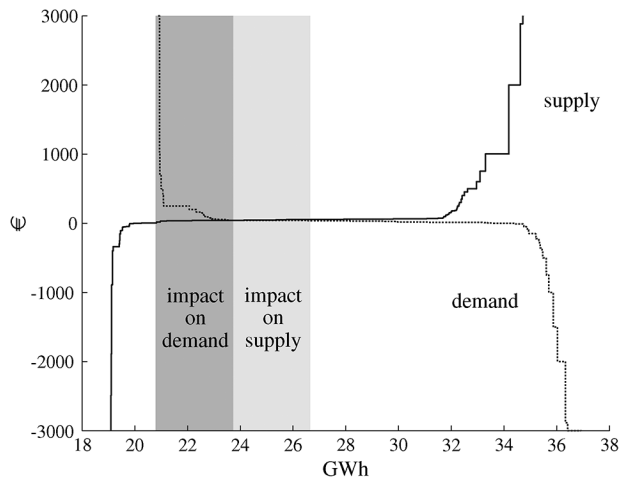


Fig. 3. Supply and demand bid curves (Wednesday September 6th, 2012, hour 24). The grey areas denote the regions where the aggregator's bids can either influence supply or demand curves.

with inflexible charging, demand is more evenly distributed throughout the day (Fig. 2(c), column 4). Inflexible demand is higher between the morning and evening and shows peaks that coincide with the price peaks in the market (Fig. 2(b), column 4). This explains the much higher costs associated with inflexible demand. The charging demand peaks are related to the commuting patterns: the first peak occurs when vehicles arrive at work and the second one when they arrive at home in the evening.

E. Computation

The simulations were run with MATLAB R2012a on a 2.67 GHz Intel Xeon processor. The optimization problems (mixed-integer linear programming (MILP) problems for the bidding strategy and linear programming problems for the benchmark cases) were solved using CPLEX v12.4. The size of the problem changes with the day being simulated, since the number of supply and demand bids to be considered changes. An overview of the MILP problem sizes and execution times is given in Table IV.

To reduce the size of the problem, we made use of the fact that the maximum aggregator bid volume at each time step is bounded, i.e., limited by the available charging power, as expressed by (4). Therefore, the bidding decisions of the aggregator can only affect a limited number of bids around the market clearing price and volume that would prevail in the absence of aggregator bids. Beyond that region, the individual demand/supply bids can be lumped in a single demand/supply bid block, reducing the number of variables and constraints in the problem. Fig. 3 illustrates the relevant regions.

IV. CONCLUSION AND DISCUSSION

In this paper a bidding strategy for a PEV aggregator bidding in the day-ahead market is introduced. The described scheme takes into account both the uncertainty in market bids and the constraints on the available demand flexibility, which are derived from a transport simulation and are also subject to uncertainty.

The main findings of the case study with market data from EEX and realistic driving patterns are outlined as follows:

- Even at low penetrations, the aggregator can have a significant impact on prices, since charging is concentrated on few hours. Consequently, the assumption of exogenous prices leads to a suboptimal bidding strategy.
- Under perfect information, the surplus maximizing dispatch and the bidding strategy lead to very similar results. This suggests that the aggregator only has limited market power potential. This market power potential could however be more important at higher PEV penetrations.
- For relatively large PEV fleets, driving uncertainty has only a minor impact on the charging costs, since aggregations of large numbers of vehicles do not exhibit large variances in the parameters describing their flexibility.
- The costs of the bidding strategy under market bid uncertainty are only slightly higher than those under perfect information, even if a simple strategy is used to estimate the other bids.
- As expected, charging costs can be significantly reduced if PEV demand is managed, compared with when demand is uncontrolled.

Note that the main reduction in charging costs respect to uncontrolled charging comes from exploiting charging flexibility in general. Although an aggregator-based centralized scheme, such as the one proposed in this paper, can potentially achieve optimal cost outcomes for the fleet as a whole, decentralized schemes, where each PEV forms its own bids, could lead to overall costs which are only a few percent higher [5]. However, although fleet-level results are comparable, the individual performance of some vehicles in a decentralized scheme could be significantly poorer than under the aggregator's management, since the decision-making related risk is no longer shared among the PEVs, and the uncertainties in driving patterns no longer balance each other. For a more detailed discussion of the advantages and disadvantages of decentralized and centralized approaches we refer to [5].

In this paper we have presented a decision-making tool for the aggregator to participate in day-ahead markets. This tool results in optimal bid volumes and prices for the day-ahead market. It could be extended to take into account other markets, and complemented by other tools focusing on other operational stages:

- Markets for ancillary services, as has been proposed in other aggregator bidding approaches [13]–[16] could be included. If the aggregator is a price-taker in these markets, offering ancillary services can be easily introduced into the proposed framework as we have shown in [29].
- The participation of the aggregator in intra-day markets could also be considered, where a receding-horizon version of the proposed framework could be used. Expected

deviations from the day-ahead market schedule could be compensated for in these markets.

- The optimal bidding tool needs to be complemented by an operational tool scheduling individual vehicles in real-time, trying to match the commitments in day-ahead or other markets as closely as possible, see [20], [24], and [30] for examples.

Moreover, the case of several aggregators behaving strategically could be explored. This would require extending the MPEC formulation in the paper to a more complex EPEC (equilibrium problem with equilibrium constraints) framework [31]–[33]. With several strategic agents, each agent faces a bilevel model analog to the one described in the paper. The problem can then be recast as an EPEC, where equilibrium constraints couple the individual problems.

APPENDIX A

CHARACTERIZATION OF VIRTUAL BATTERY

A. Aggregation of End-Use Constraints

As explained in Section II-A, the energy and power constraints of the fleet are integrated into the charging optimization as constraints on a virtual battery. To find the power and energy bounds of the virtual battery the individual constraints of each PEV have to be derived first. They depend on the physical characteristics of the battery, such as charging efficiency and energy capacity, on the available charging power, as well as on the use of the vehicle. It is assumed that PEVs are plugged in if they are parked for at least one time period, i.e., one hour, no matter the location.

For each PEV v_l the upper and lower bounds of the charging power $P_{v_l}^t$ and the energy content $E_{v_l}^t$ at each time step t are calculated.

The upper bound on the energy content is calculated assuming charging starts as soon as the vehicle parks, while the lower bound is calculated assuming charging is deferred as much as possible, given that the vehicle should depart with enough energy in the battery for the forthcoming trip. The lower bounds are calculated with foresight, e.g. when there are two consecutive trips with only a short parking break between them, this is considered when computing the required charging during the parking break preceding the two trips. We assume that the energy content should not drop below 20% of the battery capacity to reduce battery degradation. It is also assumed that batteries reach a full state of charge at some point, and their energy content can therefore not drop by more than the total daily energy consumption from that reference.

Finally, the maximum charging power is primarily determined by the assumed available charging power given by the charging infrastructure or the vehicle battery. The minimum charging power in the reference case will be either zero, or whatever is necessary to satisfy the energy requirements (non deferrable or inflexible charging). It is important to take into account energy constraints when defining the available power to avoid large deviations between the simulations at aggregated and individual level.

Once the individual bounds have been defined, they contribute to the bounds of the virtual battery if the vehicle is connected ($u_{v_l} = 1$), for example for the upper energy bound:

$$E_{\max}^t = \sum_{v_l} E_{v_l, \max}^t \cdot u_{v_l}^t. \quad (35)$$

This can be done similarly for the bounds on P_a and the lower energy bound $E_{\min}^t$.

Finally the energy contributions due to arrivals and departures can be calculated as

$$E_{\text{dep}}^t = \sum_{v_l} E_{v_l, \max}^t \cdot u_{v_l}^{t-1} (u_{v_l}^{t-1} - u_{v_l}^t) \quad (36)$$

$$E_{\text{arr}}^t = \sum_{v_l} E_{v_l, \max}^t \cdot u_{v_l}^t (u_{v_l}^t - u_{v_l}^{t-1}). \quad 37$$

B. Modeling Uncertainty in Driving Patterns

From the agent-based transport simulation MATSim, we obtain the arrival and departure time of each trip and the associated energy consumption for each vehicle. Based on that reference sample, we generate different samples of driving patterns for each vehicle. We assume that the order of performed activities (e.g., home-work-shop-home) given by the transport simulations is deterministic, but we model departure times, trip durations and trip energy as stochastic variables. Specifically, we assume that the trip departure time and trip duration are uniformly distributed ± 30 min around their reference values, except when activities are within 60 min of each other. In such cases, in order to preserve the order of the activities, we reduce the support of the distribution. The trip consumption is also assumed to be uniformly distributed ± 1 kWh around the reference value. The support of this distribution is reduced in the case of short trips where negative consumption values could be generated. With these distributions, we generate different realizations of driving patterns for each individual vehicle. We then use these realizations to obtain different samples of the aggregated variables, as explained in the previous subsection. The chosen distributions are just exemplary distributions, in practice the aggregator would need to collect data from its PEV customers to find appropriate models for their stochastic behavior. However, the general approach remains valid for any distribution, since the method used to solve the probabilistic constraints does not require any particular distribution assumptions.

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